PRACTICAL : 01

Aim:

Set up Python environment for AI and implement a simple reflex or goal-based agent in a toy environment.

Theory:

A simple reflex agent selects an action based only on the current state of the environment. It uses predefined condition-action rules. In this program, the agent observes whether a location is dirty or clean. If it is dirty, the agent cleans it; otherwise, it moves to the next location.

Program:

```
print("Welcome to my first AI Robot")


room = ['D', 'C', 'D', 'C']


print("Room before cleaning:", room)


def clean_room(room):
    for i in range(len(room)):
        if room[i] == 'D':
            print(f"Cleaning room at position {i + 1}")
            room[i] = 'C'
        else:
            print(f"Room at position {i + 1} is already clean")
```

```
    return room


room = clean_room(room)


print("Room after cleaning:", room)
print("All rooms are clean!")
```

Output:

```
Welcome to my first AI Robot
Room before cleaning: ['D', 'C', 'D', 'C']
Cleaning room at position 1
Room at position 2 is already clean
Cleaning room at position 3
Room at position 4 is already clean
Room after cleaning: ['C', 'C', 'C', 'C']
All rooms are clean!
```

Conclusion:

Thus, a simple reflex agent was successfully implemented in Python. The agent correctly perceived the environment and performed actions based on predefined condition-action rules.